

Sri Lanka Institute of Information Technology



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Practical Answer Submission

Practical Sheet No: 09

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Activity 1

Unix directory structure

1.

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ pwd  
/home/kamshiga  
kamshiga@kamshiga-VirtualBox:~$
```

Shows the current working directory (absolute path from the root directory (/) to the current location where the user is working)

2.

```
kamshiga@kamshiga-VirtualBox:~$ cd /  
kamshiga@kamshiga-VirtualBox:/$ ls -l  
total 4009052  
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin  
drwxr-xr-x 2 root root 4096 Feb 26 2024 bin.usr-is-merged  
drwxr-xr-x 3 root root 4096 Mar 25 19:49 boot  
dr-xr-xr-x 2 root root 4096 Apr 24 2024 cdrom  
drwxr-xr-x 19 root root 4280 Apr 5 21:08 dev  
drwxr-xr-x 138 root root 12288 Mar 25 19:50 etc  
drwxr-xr-x 3 root root 4096 Jan 31 12:48 home  
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib  
lrwxrwxrwx 1 root root 9 Apr 22 2024 lib64 -> usr/lib64  
drwxr-xr-x 2 root root 4096 Apr 8 2024 lib.usr-is-merged  
drwx----- 2 root root 16384 Jan 31 11:14 lost+found  
drwxr-xr-x 2 root root 4096 Apr 24 2024 media  
drwxr-xr-x 2 root root 4096 Apr 24 2024 mnt  
drwxr-xr-x 2 root root 4096 Apr 24 2024 opt  
dr-xr-xr-x 288 root root 0 Apr 5 21:08 proc  
drwx----- 5 root root 4096 Jan 31 12:48 root  
drwxr-xr-x 34 root root 840 Apr 5 21:08 run  
lrwxrwxrwx 1 root root 8 Apr 22 2024 sbin -> usr/sbin  
drwxr-xr-x 2 root root 4096 Mar 31 2024 sbin.usr-is-merged  
drwxr-xr-x 15 root root 4096 Mar 21 11:33 snap  
drwxr-xr-x 2 root root 4096 Apr 24 2024 srv  
-rw----- 1 root root 4105175040 Jan 31 12:10 swap.img  
dr-xr-xr-x 13 root root 0 Apr 5 21:08 sys  
drwxrwxrwt 17 root root 4096 Apr 5 21:09 tmp  
drwxr-xr-x 12 root root 4096 Apr 24 2024 usr  
drwxr-xr-x 14 root root 4096 Jan 31 12:47 var  
kamshiga@kamshiga-VirtualBox:/$
```

When I use the `ls -l` command, it shows detailed information about files and directories. The first character in each line indicates the type of file. For example, `d` represents a directory, `-` represents a normal file, and `l` represents a symbolic link. In special directories like `/dev`, `c` and `b` are used to represent character and block devices.

The next nine characters show the permissions for the owner, group, and others. These include r for read, w for write, and x for execute. For instance, rwxr-xr-x means the owner has full permissions, while the group and others have only read and execute permissions.

After the permissions, the output also includes additional details such as the number of links to the file, the owner name, the group name, the file size in bytes, the last modified date and time, and finally the file or directory name.

3.

```
kamshiga@kamshiga-VirtualBox:/$ cd /bin
kamshiga@kamshiga-VirtualBox:/bin$ ls -l
total 224024
-rwxr-xr-x 1 root root 55744 Apr 5 2024 '['
-rwxr-xr-x 1 root root 18744 Mar 31 2024 aa-enabled
-rwxr-xr-x 1 root root 18744 Mar 31 2024 aa-exec
-rwxr-xr-x 1 root root 18736 Mar 31 2024 aa-features-abi
-rwxr-xr-x 1 root root 22912 Apr 7 2024 aconnect
-rwxr-xr-x 1 root root 1622 Mar 6 06:13 acpidbg
-rwxr-xr-x 1 root root 16422 Apr 18 2024 add-apt-repository
-rwxr-xr-x 1 root root 14720 Mar 6 21:30 addpart
lrwxrwxrwx 1 root root 26 Dec 3 20:31 addr2line -> x86_64-linux-gnu-addr2line
-rwxr-xr-x 1 root root 158576 Apr 8 2024 aircan-discover
-rwxr-xr-x 1 root root 47552 Apr 7 2024 alsabat
-rwxr-xr-x 1 root root 85328 Apr 7 2024 alsaloop
-rwxr-xr-x 1 root root 82216 Apr 7 2024 alsamixer
-rwxr-xr-x 1 root root 76160 Apr 7 2024 alsatplg
-rwxr-xr-x 1 root root 31576 Apr 7 2024 alsaucm
-rwxr-xr-x 1 root root 31112 Apr 7 2024 amidi
-rwxr-xr-x 1 root root 63952 Apr 7 2024 amixer
-rwxr-xr-x 1 root root 274 Apr 8 2024 apg
-rwxr-xr-x 1 root root 26784 Apr 8 2024 apgbfm
-rwxr-xr-x 1 root root 84400 Apr 7 2024 aplay
-rwxr-xr-x 1 root root 27016 Apr 7 2024 aplaymidi
-rwxr-xr-x 1 root root 2322 Apr 18 2024 apport-bug
-rwxr-xr-x 1 root root 13625 Jul 8 2025 apport-cli
lrwxrwxrwx 1 root root 10 Jul 8 2025 apport-collect -> apport-bug
-rwxr-xr-x 1 root root 3790 Jul 8 2025 apport-unpack
-rwxr-xr-x 1 root root 14648 Mar 31 2024 appres
-rwxr-xr-x 1 root root 141544 Apr 8 2024 appstreamcli
lrwxrwxrwx 1 root root 6 Apr 8 2024 apropos -> whatis
-rwxr-xr-x 1 root root 18824 Mar 31 2024 ar
```

```
kamshiga@kamshiga-VirtualBox:/$ cd /home
kamshiga@kamshiga-VirtualBox:/home$ ls -l
total 4
drwxr-x-- 16 kamshiga kamshiga 4096 Mar 26 16:47 kamshiga
```

```
kamshiga@kamshiga-VirtualBox:/home$ cd /dev
kamshiga@kamshiga-VirtualBox:/dev$ ls -l
total 0
crw-r--r-- 1 root root 10, 235 Apr 5 21:08 autofs
drwxr-xr-x 2 root root 520 Apr 5 21:13 block
drwxr-xr-x 2 root root 80 Apr 5 21:08 bsg
crw----- 1 root root 10, 234 Apr 5 21:08 btrfs-control
drwxr-xr-x 3 root root 60 Apr 5 21:08 bus
lrwxrwxrwx 1 root root 3 Apr 5 21:08 cdrom -> sr0
drwxr-xr-x 2 root root 3740 Apr 5 21:08 char
crw----- 1 root root 5, 1 Apr 5 21:08 console
lrwxrwxrwx 1 root root 11 Apr 5 21:08 core -> /proc/kcore
drwxr-xr-x 5 root root 100 Apr 5 21:08 cpu
crw----- 1 root root 10, 260 Apr 5 21:08 cpu_dma_latency
crw----- 1 root root 10, 203 Apr 5 21:08 cuse
drwxr-xr-x 9 root root 180 Apr 5 21:08 disk
drwxr-xr-x 2 root root 60 Apr 5 21:08 dma_heap
drwxr-xr-x 3 root root 100 Apr 5 21:08 dri
crw----- 1 root root 10, 258 Apr 5 21:08 ecryptfs
crw-rw---- 1 root video 29, 0 Apr 5 21:08 fb0
lrwxrwxrwx 1 root root 13 Apr 5 21:08 fd -> /proc/self/fd
crw-rw-rw- 1 root root 1, 7 Apr 5 21:08 full
crw-rw-rw- 1 root root 10, 229 Apr 5 21:08 fuse
crw----- 1 root root 240, 0 Apr 5 21:08 hidraw0
crw----- 1 root root 10, 228 Apr 5 21:08 hpet
drwxr-xr-x 2 root root 0 Apr 5 21:08 hugepages
crw----- 1 root root 10, 183 Apr 5 21:08 hwrng
crw----- 1 root root 89, 0 Apr 5 21:08 i2c-0
lrwxrwxrwx 1 root root 12 Apr 5 21:08 initctl -> /run/initctl
drwxr-xr-x 4 root root 340 Apr 5 21:08 input
```

```

kamshiga@kamshiga-VirtualBox:/dev$ cd /etc
kamshiga@kamshiga-VirtualBox:/etc$ ls -l
total 1124
-rw-r--r-- 1 root root 3444 Jul 5 2023 adduser.conf
drwxr-xr-x 3 root root 4096 Apr 24 2024 alsa
drwxr-xr-x 2 root root 4096 Jan 31 18:56 alternatives
-rw-r--r-- 1 root root 335 Apr 8 2024 anacrontab
-rw-r--r-- 1 root root 433 Apr 8 2024 apg.conf
drwxr-xr-x 5 root root 4096 Apr 24 2024 apm
drwxr-xr-x 2 root root 4096 Apr 24 2024 apparmor
drwxr-xr-x 9 root root 4096 Mar 25 19:50 apparmor.d
drwxr-xr-x 3 root root 4096 Jan 31 12:42 appport
drwxr-xr-x 9 root root 4096 Jan 31 11:10 apt
drwxr-xr-x 3 root root 4096 Jan 31 12:41 avahi
-rw-r--r-- 1 root root 2319 Mar 31 2024 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 25 2020 bash_completion
-rw-r--r-- 1 root root 367 Aug 2 2022 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 19 2024 binfmt.d
drwxr-xr-x 2 root root 4096 Apr 24 2024 bluetooth
-rw-r--r-- 1 root root 33 Apr 24 2024 brlapi.key
drwxr-xr-x 7 root root 4096 Apr 24 2024 brltty
-rw-r--r-- 1 root root 30571 Mar 31 2024 brltty.conf
drwxr-xr-x 3 root root 4096 Apr 24 2024 ca-certificates
-rw-r--r-- 1 root root 6288 Apr 24 2024 ca-certificates.conf
drwxr-s-- 2 root dip 4096 Apr 24 2024 chatscripts
drwxr-xr-x 5 root root 4096 Jan 31 12:49 cloud
drwxr-xr-x 2 colord 4096 Jan 31 12:48 colord
drwxr-xr-x 2 root 4096 Jan 31 11:22 console-setup
drwxr-xr-x 2 root 4096 Apr 24 2024 cracklib
drwx----- 2 root root 4096 Apr 19 2024 credstore
drwx----- 2 root root 4096 Apr 19 2024 credstore.encrypted

```

```

kamshiga@kamshiga-VirtualBox:/etc$ cd /lib
kamshiga@kamshiga-VirtualBox:/lib$ ls -l
total 640
drwxr-xr-x 2 root root 4096 Apr 24 2024 app
drwxr-xr-x 2 root root 4096 Apr 24 2024 apparmor
drwxr-xr-x 5 root root 4096 Apr 24 2024 apt
drwxr-xr-x 3 root root 12288 Apr 24 2024 aspell
drwxr-xr-x 2 root root 4096 Jan 31 18:56 bfd-plugins
drwxr-xr-x 2 root root 4096 Mar 25 19:46 binfmt.d
drwxr-xr-x 2 root root 4096 Apr 24 2024 brltty
drwxr-xr-x 2 root root 4096 Jan 31 12:41 cloud-init
-rwxr-xr-x 1 root root 1075 Jan 20 2023 cnf-update-db
-rwxr-xr-x 1 root root 3565 Jan 20 2023 command-not-found
drwxr-xr-x 2 root root 4096 Jan 31 18:56 compat-ld
drwxr-xr-x 2 root root 4096 Apr 24 2024 console-setup
lrwxrwxrwx 1 root root 21 Jan 31 2024 cpp -> /etc/alternatives/cpp
drwxr-xr-x 10 root root 4096 Apr 24 2024 cups
drwxr-xr-x 2 root root 4096 Apr 24 2024 dbus-1.0
drwxr-xr-x 3 root root 4096 Apr 24 2024 debug
drwxr-xr-x 3 root root 4096 Apr 24 2024 dhcpcd
drwxr-xr-x 3 root root 4096 Apr 24 2024 dpkg
drwxr-xr-x 2 root root 4096 Apr 24 2024 dracut
drwxr-xr-x 3 root root 4096 Apr 24 2024 emacs-common
drwxr-xr-x 2 root root 4096 Mar 25 19:50 environment.d
drwxr-xr-x 7 root root 4096 Apr 24 2024 evolution-data-server
drwxr-xr-x 2 root root 4096 Apr 24 2024 file
drwxr-xr-x 3 root root 4096 Apr 24 2024 firewallld
drwxr-xr-x 100 root root 28672 Mar 25 19:38 firmware
drwxr-xr-x 3 root root 4096 Jan 31 18:56 gcc

```

When I use the `ls -l` command, it displays detailed information about files and directories in a list format. The first character in each line shows the type of file. For example, `d` indicates a directory, `-` indicates a regular file, and `l` represents a symbolic link. In system directories such as `/dev`, the letters `c` and `b` are used to represent character and block device files.

The next nine characters represent the file permissions for the owner, group, and others. These include `r` for read, `w` for write, and `x` for execute permissions. For example, `rwxr-xr-x` means the owner has full permissions, while the group and others have only read and execute access.

In addition to permissions, the output also provides other details such as the number of links to the file, the owner name, the group name, the file size in bytes, the date and time of last modification, and the name of the file or directory.

4.

```
kamshiga@kamshiga-VirtualBox:~$ cd  
kamshiga@kamshiga-VirtualBox:~$ cd ~  
kamshiga@kamshiga-VirtualBox:~$ cd $HOME  
kamshiga@kamshiga-VirtualBox:~$
```

```
kamshiga@kamshiga-VirtualBox:~$ pwd  
/home/kamshiga  
kamshiga@kamshiga-VirtualBox:~$ cd .  
kamshiga@kamshiga-VirtualBox:~$ pwd  
/home/kamshiga  
kamshiga@kamshiga-VirtualBox:~$ cd ..  
kamshiga@kamshiga-VirtualBox:/home$ pwd  
/home  
kamshiga@kamshiga-VirtualBox:/home$
```

The `cd` command is used to change directories in the file system.

- `cd -` - takes me directly to my home directory.
- `cd ~` - also navigates me to my home directory, since `~` represents the home path.
- `cd $HOME` - does the same, because `$HOME` is an environment variable that stores the path of my home directory. The difference among `cd`, `cd .`, and `cd ..` is:
 - `cd -` - moves me to my home directory.
 - `cd .` - keeps the user in the current directory (‘ . ’ refers to the current location).
 - `cd ..` - moves one level up to the parent directory (‘ .. ’ refers to the parent folder).

These commands are useful for navigating through directories efficiently and managing file locations within the system.

Activity 2

File Access permissions

1.

```
kamshiga@kamshiga-VirtualBox:~$ cd ~
kamshiga@kamshiga-VirtualBox:~$ cd $HOME
kamshiga@kamshiga-VirtualBox:~$ mkdir IT24101365
kamshiga@kamshiga-VirtualBox:~$ cd IT24101365
kamshiga@kamshiga-VirtualBox:~/IT24101365$ touch chkpermission
kamshiga@kamshiga-VirtualBox:~/IT24101365$ ls -l chkpermission
-rw-rw-r-- 1 kamshiga kamshiga 0 Apr  5 21:31 chkpermission
kamshiga@kamshiga-VirtualBox:~/IT24101365$
```

When new file created, it is assigned default permissions based on system settings.

2.

```
kamshiga@kamshiga-VirtualBox:~/IT24101365$ chmod 700 chkpermission
kamshiga@kamshiga-VirtualBox:~/IT24101365$ ls -l chkpermission
-rwx----- 1 kamshiga kamshiga 0 Apr  5 21:31 chkpermission
kamshiga@kamshiga-VirtualBox:~/IT24101365$
```

By modifying file permissions, access control can be enforced. Giving full permissions to the owner and removing permissions from others ensures that only the owner can read, write, and execute the file.

3.

```
kamshiga@kamshiga-VirtualBox:~/IT24101365$ chmod 755 chkpermission
kamshiga@kamshiga-VirtualBox:~/IT24101365$ ls -l chkpermission
-rwxr-xr-x 1 kamshiga kamshiga 0 Apr  5 21:31 chkpermission
kamshiga@kamshiga-VirtualBox:~/IT24101365$
```

Permissions can be adjusted to allow limited access to other users. In this case, the owner has full control, while group and others are allowed to read and execute the file but cannot modify it.

4.

```
kamshiga@kamshiga-VirtualBox:~/IT24101365$ cd ~
kamshiga@kamshiga-VirtualBox:~$ ls -ld IT24101365
drwxrwxr-x 2 kamshiga kamshiga 4096 Apr  5 21:31 IT24101365
kamshiga@kamshiga-VirtualBox:~$
```

When a directory is created, it usually has read, write, and execute permissions for the owner, and read and execute permissions for group and others. The execute permission is important for accessing directory contents.

5.

```
kamshiga@kamshiga-VirtualBox:~$ chmod 700 IT24101365
kamshiga@kamshiga-VirtualBox:~$ ls -ld IT24101365
drwx----- 2 kamshiga kamshiga 4096 Apr  5 21:31 IT24101365
kamshiga@kamshiga-VirtualBox:~$
```

Removing all permissions from others ensures that only the owner (and possibly group) can access the directory. Other users cannot read, write, or enter the directory.

6.

```
kamshiga@kamshiga-VirtualBox:~$ chmod u-rx IT24101365
kamshiga@kamshiga-VirtualBox:~$ ls -ld IT24101365
d-w----- 2 kamshiga kamshiga 4096 Apr  5 21:31 IT24101365
kamshiga@kamshiga-VirtualBox:~$
```

When read and execute permissions are removed from the owner, the owner cannot view or access the directory contents. This results in permission denied errors when trying to list files.

7.

```
kamshiga@kamshiga-VirtualBox:~$ chmod u+r IT24101365
kamshiga@kamshiga-VirtualBox:~$ ls -ld IT24101365
drw----- 2 kamshiga kamshiga 4096 Apr  5 21:31 IT24101365
kamshiga@kamshiga-VirtualBox:~$
```

Giving only read permission is not sufficient to access a directory. Without execute permission, the user cannot enter or properly view the directory contents.

8.

```
kamshiga@kamshiga-VirtualBox:~$ chmod u+rw IT24101365
kamshiga@kamshiga-VirtualBox:~$ ls -ld IT24101365
drw----- 2 kamshiga kamshiga 4096 Apr  5 21:31 IT24101365
kamshiga@kamshiga-VirtualBox:~$
```

When both read and execute permissions are given, the user can successfully access and list the contents of the directory. Execute permission allows entering the directory, while read permission allows viewing file names.

Activity 3

Changing file/directory modes

1.

```
kamshiga@kamshiga-VirtualBox:~$ mkdir lab1
kamshiga@kamshiga-VirtualBox:~$ cd lab1
```

2.

```
kamshiga@kamshiga-VirtualBox:~/lab1$ nano idioms
kamshiga@kamshiga-VirtualBox:~/lab1$ ls -l idioms
-rw-rw-r-- 1 kamshiga kamshiga 189 Apr  5 21:52 idioms
kamshiga@kamshiga-VirtualBox:~/lab1$
```

```
GNU nano 7.2 idioms *
Try, Try, Try, One day you can fly
Ideas without action are worthless
What doesn't kill me makes me stronger
Patience is the companion of wisdom
Only those who do nothing make no mistakes
```

```
kamshiga@kamshiga-VirtualBox:~/lab1$ chmod a+rwX idioms
kamshiga@kamshiga-VirtualBox:~/lab1$ ls -l idioms
-rwxrwxrwx 1 kamshiga kamshiga 189 Apr  5 21:52 idioms
kamshiga@kamshiga-VirtualBox:~/lab1$
```

```
kamshiga@kamshiga-VirtualBox:~/lab1$ chmod 644 idioms
kamshiga@kamshiga-VirtualBox:~/lab1$ ls -l idioms
-rw-r--r-- 1 kamshiga kamshiga 189 Apr  5 21:52 idioms
kamshiga@kamshiga-VirtualBox:~/lab1$
```

The `chmod` command is used to modify file permissions. Using symbolic mode (like `a+rwX`) grants permissions to all users, while numeric mode (like `644`) sets specific permission levels for owner, group, and others.

3.

```
kamshiga@kamshiga-VirtualBox:~/lab1$ ls -li
263338 idioms
kamshiga@kamshiga-VirtualBox:~/lab1$
```

The `ls -li` command is used to display the i-node number of a file. The i-node uniquely identifies a file in the file system.

4.

```
kamshiga@kamshiga-VirtualBox:~/lab1$ ls -l idioms
-rw-r--r-- 1 kamshiga kamshiga 189 Apr  5 21:52 idioms
kamshiga@kamshiga-VirtualBox:~/lab1$
```

The link count shows how many hard links are associated with a file. Initially, a file has one link, but creating hard links increases this count.

Activity 4

Hard links and soft links

1.

```
kamshiga@kamshiga-VirtualBox:~/lab1$ cd ~
kamshiga@kamshiga-VirtualBox:~$ mkdir linkdemo
kamshiga@kamshiga-VirtualBox:~$ cd linkdemo
kamshiga@kamshiga-VirtualBox:~/linkdemo$ mkdir softln hardln
kamshiga@kamshiga-VirtualBox:~/linkdemo$
```

Create directories to organize files and demonstrate linking concepts. Separate folders are used for hard links and soft links.

2.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo$ cd hardln
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ ln ~/lab1/idioms hardidioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$
```

A hard link is created as another name for the same file. It points directly to the same i-node, meaning both files share the same data.

3.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ ls -l hardidioms ~/lab1/idioms
-rw-r--r-- 2 kamshiga kamshiga 189 Apr  5 21:52 hardidioms
-rw-r--r-- 2 kamshiga kamshiga 189 Apr  5 21:52 /home/kamshiga/lab1/idioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$
```

When checking properties, hard-linked files have the same i-node number and file size. The link count increases as more hard links are created.

4.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ nano ~/lab1/idioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ cat hardidioms
Try, Try, Try, One day you can fly
Ideas without action are worthless
What doesn't kill me makes me stronger
Patience is the companion of wisdom
Only those who do nothing make no mistakes
Knowledge is power
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$
```

When the original file is edited, the changes are reflected in the hard-linked file as well. This is because both names refer to the same physical data on disk.

5.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ ln ~/lab1/idioms hardidioms_2
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ ls -l hardidioms ~/lab1/idioms hardidioms_2
-rw-r--r-- 3 kamshiga kamshiga 207 Apr  5 22:24 hardidioms
-rw-r--r-- 3 kamshiga kamshiga 207 Apr  5 22:24 hardidioms
-rw-r--r-- 3 kamshiga kamshiga 207 Apr  5 22:24 hardidioms_2
-rw-r--r-- 3 kamshiga kamshiga 207 Apr  5 22:24 /home/kamshiga/lab1/idioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$
```

Creating multiple hard links increases the link count. Deleting one link does not remove the file as long as other links exist.

Hard Link Observation:

- Hard links share the same i-node and data. Any modification in one file affects all linked files. The file is only deleted when all links are removed.

6.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo/hardln$ cd ../softln
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$ ln -s ~/lab1/idioms softidioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$
```

A soft link (symbolic link) is created as a reference to another file. It stores the path of the original file rather than pointing directly to its data.

7.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$ ls -l softidioms ~/lab1/idioms
-rw-r--r-- 3 kamshiga kamshiga 207 Apr  5 22:24 /home/kamshiga/lab1/idioms
lrwxrwxrwx 1 kamshiga kamshiga 26 Apr  5 22:28 softidioms -> /home/kamshiga/lab1/idioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$ ls -li softidioms ~/lab1/idioms
263338 /home/kamshiga/lab1/idioms 263323 softidioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$
```

Soft links have a different i-node number from the original file. The link count of the original file does not change when a soft link is created.

8.

```
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$ ls -l softtdioms ~/lab1/idioms
-rw-r--r-- 3 kamshiga kamshiga 207 Apr  5 22:24 /home/kamshiga/lab1/idioms
lrwxrwxrwx 1 kamshiga kamshiga 26 Apr  5 22:28 softtdioms -> /home/kamshiga/lab1/idioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$ ls -i softtdioms ~/lab1/idioms
263338 /home/kamshiga/lab1/idioms 263323 softtdioms
kamshiga@kamshiga-VirtualBox:~/linkdemo/softln$
```

Changes made to the original file are visible through the soft link. However, if the original file is deleted, the soft link becomes invalid.

9.

```
kamshiga@kamshiga-VirtualBox:~$ ln -s ~/lab1/idioms softtdioms_2
kamshiga@kamshiga-VirtualBox:~$ ls -l ~/softtdioms_2
lrwxrwxrwx 1 kamshiga kamshiga 26 Apr  5 22:39 /home/kamshiga/softtdioms_2 -> /home/kamshiga/lab1/idioms
kamshiga@kamshiga-VirtualBox:~$ ls -i ~/softtdioms_2
263343 /home/kamshiga/softtdioms_2
kamshiga@kamshiga-VirtualBox:~$
```

Final Observation:

- Soft links work like shortcuts to the original file and depend on it. They have their own i-node numbers, which are different from the original file. If the original file is deleted, the soft link will no longer work. However, deleting the soft link does not affect the original file.